CustomerSegmentation_Project

Segment customers by purchase behaviour (RFM) and preferences using **Python, scikit-learn (KMeans)**.

Dataset

`Sales_Data.csv` - 2,000 order lines, 400 customers (2023-2025). Sample dataset (Superstore-style) with Customer ID, Segment, Region, Category, Sales, Discount, Profit.

Method

1. Aggregated orders per customer: **Recency, Frequency, Monetary, Avg Discount**
2. Log-transform + StandardScaler
3. Chose k using Elbow + Silhouette, final **KMeans (k=4)**
4. Named segments from cluster profiles, visualised with PCA and bar charts

Segments

Segment	Customers	Avg Recency (days)	Avg Orders	Avg Spend	Revenue
Loyal Customers	164	247	5.3	$7,948	$1.30M
Champions (High Value)	112	42	6.8	$8,041	$0.90M
At-Risk / Lost	71	316	3.3	$2,206	$0.16M
Occasional / New	48	306	2.7	$943	$0.05M

Key Insights

- Champions and Loyal customers (69% of customers) generate ~91% of revenue.
- Loyal customers spend as much as Champions but have not bought recently: a re-engagement opportunity.
- At-Risk customers get the highest discounts (18%) yet spend the least and give the

lowest profit: stop blanket discounting.

- Occasional/New customers have low spend: use welcome offers and cross-sell.

Recommendations

- Champions: loyalty rewards, early access
- Loyal: win-back emails / reminders
- At-Risk: targeted, limited offers instead of deep discounts
- Occasional/New: onboarding offers, bundles

Files

`customer_segmentation.py` (code), `customers_segmented.csv`, `segment_summary.csv`, charts `1-5*.png`

Run

`pip install pandas scikit-learn matplotlib seaborn` then `python customer_segmentation.py`